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CS 323: Database Systems / Database Management

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Question 1

```
MariaDB [(none)]> CREATE DATABASE elitevideo;  
Query OK, 1 row affected (0.024 sec)
```

```
MariaDB [(none)]> USE elitevideo;  
Database changed
```

```
MariaDB [elitevideo]> CREATE TABLE MEMBERSHIP (Mem_Num INT PRIMARY  
KEY, Mem_FName VARCHAR(50), Mem_LName VARCHAR(50), Mem_Street  
VARCHAR(150), Mem_City VARCHAR(50), Mem_State CHAR(2), Mem_Zip  
VARCHAR(10), Mem_Balance DECIMAL(6,2));  
Query OK, 0 rows affected (0.037 sec)
```

```
MariaDB [elitevideo]> CREATE TABLE RENTAL ( Rent_Num INT PRIMARY  
KEY, Rent_Date DATE, Mem_Num INT NOT NULL, FOREIGN KEY (Mem_Num)  
REFERENCES MEMBERSHIP(Mem_Num));  
Query OK, 0 rows affected (0.023 sec)
```

```
MariaDB [elitevideo]> CREATE TABLE PRICE ( Price_Code INT PRIMARY  
KEY, Price_Description VARCHAR(90), Price_RentFee DECIMAL(5,2),  
Price_DailyLateFee DECIMAL(5,2));  
Query OK, 0 rows affected (0.025 sec)
```

```
MariaDB [elitevideo]> CREATE TABLE MOVIE ( Movie_Num INT PRIMARY  
KEY, Movie_Name VARCHAR(100) NOT NULL, Movie_Year INT, Movie_Cost  
DECIMAL(6,2), Movie_Genre VARCHAR(90), Price_Code INT, FOREIGN KEY  
(Price_Code) REFERENCES PRICE(Price_Code));  
Query OK, 0 rows affected (0.020 sec)
```

```
MariaDB [elitevideo]> CREATE TABLE VIDEO (Vid_Num INT PRIMARY KEY,  
Vid_InDate DATE, Movie_Num INT NOT NULL, FOREIGN KEY (Movie_Num)  
REFERENCES MOVIE(Movie_Num));  
Query OK, 0 rows affected (0.024 sec)
```

```
MariaDB [elitevideo]> CREATE TABLE DETAILRENTAL (  
Rent_Num INT, Vid_Num INT, Detail_Fee DECIMAL(5,2), Detail_DueDate  
DATE, Detail_ReturnDate DATE, Detail_DailyLateFee DECIMAL(5,2), PRIMARY  
KEY (Rent_Num, Vid_Num), FOREIGN KEY (Rent_Num) REFERENCES  
RENTAL(Rent_Num), FOREIGN KEY (Vid_Num) REFERENCES VIDEO(Vid_Num));  
Query OK, 0 rows affected (0.025 sec)
```

Question 2

```
INSERT INTO PRICE VALUES (1, 'Standard', 2, 1);  
INSERT INTO PRICE VALUES (2, 'New Release', 3.5, 3);  
INSERT INTO PRICE VALUES (3, 'Discount', 1.5, 1);
```

INSERT INTO PRICE VALUES (4, 'Weekly Special', 1, 0.5);

INSERT INTO MEMBERSHIP VALUES (102, 'Tami', 'Dawson', '2632 Takli Circle', 'Norene', 'TN', '37136', 11);

INSERT INTO MEMBERSHIP VALUES (103, 'Curt', 'Knight', '4025 Cornell Court', 'Flatgap', 'KY', '41219', 6);

INSERT INTO MEMBERSHIP VALUES (104, 'Jamal', 'Melendez', '788 East 145th Avenue', 'Quebeck', 'TN', '38579', 0);

INSERT INTO MEMBERSHIP VALUES (105, 'Iva', 'Mcclain', '6045 Musket Ball Circle', 'Summit', 'KY', '42783', 15);

INSERT INTO MEMBERSHIP VALUES (106, 'Miranda', 'Parks', '4469 Maxwell Place', 'Germantown', 'TN', '38183', 0);

INSERT INTO MEMBERSHIP VALUES (107, 'Rosario', 'Elliott', '7578 Danner Avenue', 'Columbia', 'TN', '38402', 5);

INSERT INTO MEMBERSHIP VALUES (108, 'Mattie', 'Guy', '4390 Evergreen Street', 'Lily', 'KY', '40740', 0);

INSERT INTO MEMBERSHIP VALUES (109, 'Clint', 'Ochoa', '1711 Elm Street', 'Greeneville', 'TN', '37745', 10);

INSERT INTO MEMBERSHIP VALUES (110, 'Lewis', 'Rosales', '4524 Southwind Circle', 'Counce', 'TN', '38326', 0);

INSERT INTO MEMBERSHIP VALUES (111, 'Stacy', 'Mann', '2789 East Cook Avenue', 'Murfreesboro', 'TN', '37132', 8);

INSERT INTO MEMBERSHIP VALUES (112, 'Luis', 'Trujillo', '7267 Melvin Avenue', 'Heiskell', 'TN', '37754', 3);

INSERT INTO MEMBERSHIP VALUES (113, 'Minnie', 'Gonzales', '6430 Vasili Drive', 'Williston', 'TN', '38076', 0);

INSERT INTO MOVIE VALUES (1234, 'The Cesar Family Christmas', 2014, 39.95, 'FAMILY', 2);

INSERT INTO MOVIE VALUES (1235, 'Smokey Mountain Wildlife', 2011, 59.95, 'ACTION', 1);

INSERT INTO MOVIE VALUES (1236, 'Richard Goodhope', 2015, 59.95, 'DRAMA', 2);

```
INSERT INTO MOVIE VALUES (1237, 'Beatnik Fever', 2014, 29.95,
'COMEDY', 2);

INSERT INTO MOVIE VALUES (1238, 'Constant Companion', 2015,
89.95, 'DRAMA', NULL);

INSERT INTO MOVIE VALUES (1239, 'Where Hope Dies', 2005,
25.49, 'DRAMA', 3);

INSERT INTO MOVIE VALUES (1245, 'Time to Burn', 2012, 45.49,
'ACTION', 1);

INSERT INTO MOVIE VALUES (1246, 'What He Doesn\'t Know', 2013,
58.29, 'COMEDY', 1);
```

```
INSERT INTO VIDEO VALUES (54321, '2015-06-18', 1234);
INSERT INTO VIDEO VALUES (54324, '2015-06-18', 1234);
INSERT INTO VIDEO VALUES (54325, '2015-06-18', 1234);
INSERT INTO VIDEO VALUES (34341, '2014-01-22', 1235);
INSERT INTO VIDEO VALUES (34342, '2014-01-22', 1235);
INSERT INTO VIDEO VALUES (34366, '2016-03-02', 1236);
INSERT INTO VIDEO VALUES (34367, '2016-03-02', 1236);
INSERT INTO VIDEO VALUES (34368, '2016-03-02', 1236);
INSERT INTO VIDEO VALUES (34369, '2016-03-02', 1236);
INSERT INTO VIDEO VALUES (44392, '2015-10-21', 1237);
INSERT INTO VIDEO VALUES (44397, '2015-10-21', 1237);
INSERT INTO VIDEO VALUES (59237, '2016-02-14', 1237);
INSERT INTO VIDEO VALUES (61388, '2014-01-25', 1239);
INSERT INTO VIDEO VALUES (61353, '2013-01-28', 1245);
INSERT INTO VIDEO VALUES (61354, '2013-01-28', 1245);
INSERT INTO VIDEO VALUES (61367, '2015-07-30', 1246);
INSERT INTO VIDEO VALUES (61369, '2015-07-30', 1246);
```

```
INSERT INTO RENTAL VALUES (1001, '2016-03-01', 103);
INSERT INTO RENTAL VALUES (1002, '2016-03-01', 105);
INSERT INTO RENTAL VALUES (1003, '2016-03-02', 102);
INSERT INTO RENTAL VALUES (1004, '2016-03-02', 110);
INSERT INTO RENTAL VALUES (1005, '2016-03-02', 111);
INSERT INTO RENTAL VALUES (1006, '2016-03-02', 107);
INSERT INTO RENTAL VALUES (1007, '2016-03-02', 104);
INSERT INTO RENTAL VALUES (1008, '2016-03-03', 105);
INSERT INTO RENTAL VALUES (1009, '2016-03-03', 111);
```

```
INSERT INTO DETAILRENTAL VALUES (1001, 34342, 2, '2016-03-04',
'2016-03-02', NULL);
INSERT INTO DETAILRENTAL VALUES (1001, 61353, 2, '2016-03-04',
'2016-03-03', 1);
INSERT INTO DETAILRENTAL VALUES (1001, 34366, 3.5, '2016-03-
04', '2016-03-02', 3);
INSERT INTO DETAILRENTAL VALUES (1002, 59237, 3.5, '2016-03-
04', '2016-03-04', 3);
INSERT INTO DETAILRENTAL VALUES (1003, 54325, 3.5, '2016-03-
04', '2016-03-09', 3);
INSERT INTO DETAILRENTAL VALUES (1003, 61369, 2, '2016-03-06',
'2016-03-09', 1);
INSERT INTO DETAILRENTAL VALUES (1003, 61388, 0, '2016-03-06',
'2016-03-09', 1);
INSERT INTO DETAILRENTAL VALUES (1004, 44392, 3.5, '2016-03-
05', '2016-03-07', 3);
INSERT INTO DETAILRENTAL VALUES (1004, 34367, 3.5, '2016-03-
05', '2016-03-07', 3);
INSERT INTO DETAILRENTAL VALUES (1004, 34341, 2, '2016-03-07',
'2016-03-07', 1);
```

```
INSERT INTO DETAILRENTAL VALUES (1005, 34342, 2, '2016-03-07',
'2016-03-05', 1);
```

```
INSERT INTO DETAILRENTAL VALUES (1005, 44397, 3.5, '2016-03-
05', '2016-03-05', 3);
```

```
INSERT INTO DETAILRENTAL VALUES (1006, 34366, 3.5, '2016-03-
05', '2016-03-04', 3);
```

```
INSERT INTO DETAILRENTAL VALUES (1006, 61367, 2, '2016-03-07',
NULL, 1);
```

```
INSERT INTO DETAILRENTAL VALUES (1007, 34368, 3.5, '2016-03-
05', NULL, 3);
```

```
INSERT INTO DETAILRENTAL VALUES (1008, 34369, 3.5, '2016-03-
05', '2016-03-05', 3);
```

```
INSERT INTO DETAILRENTAL VALUES (1009, 54324, 3.5, '2016-03-
05', NULL, 3);
```

Question 3

```
MariaDB [elitevideo]> SELECT Movie_Num, Movie_Name, Movie_Year FROM
MOVIE WHERE Movie_Num = 1245;
```

Movie_Num	Movie_Name	Movie_Year
1245	Time to Burn	2012

1 row in set (0.001 sec)

Question 4

```
MariaDB [elitevideo]> SELECT Movie_Num, Movie_Name, Movie_Genre,
Price_Code FROM MOVIE WHERE Movie_Genre = 'ACTION';
```

Movie_Num	Movie_Name	Movie_Genre	Price_Code
1235	Smokey Mountain Wildlife	ACTION	1
1245	Time to Burn	ACTION	1

2 rows in set (0.001 sec)

```
MariaDB [elitevideo]> UPDATE MOVIE
```

```
-> SET Price_Code = 3
```

```
-> WHERE Movie_Genre = 'ACTION';
```

Query OK, 2 rows affected (0.003 sec)

Rows matched: 2 Changed: 2 Warnings: 0

```
MariaDB [elitevideo]> SELECT Movie_Num, Movie_Name, Movie_Genre,
Price_Code FROM MOVIE WHERE Movie_Genre = 'ACTION';
```

Movie_Num	Movie_Name	Movie_Genre	Price_Code
-----------	------------	-------------	------------

Movie_Num	Movie_Name	Movie_Genre	Price_Code
1235	Smokey Mountain Wildlife	ACTION	3
1245	Time to Burn	ACTION	3

2 rows in set (0.001 sec)

Question 5

MariaDB [elitevideo]> SELECT * FROM PRICE;

Price_Code	Price_Description	Price_RentFee	Price_DailyLateFee
1	Standard	2.00	1.00
2	New Release	3.50	3.00
3	Discount	1.50	1.00
4	Weekly Special	1.00	0.50

4 rows in set (0.001 sec)

MariaDB [elitevideo]> UPDATE PRICE SET Price_RentFee = Price_RentFee + 0.50;

Query OK, 4 rows affected (0.003 sec)

Rows matched: 4 Changed: 4 Warnings: 0

MariaDB [elitevideo]> SELECT * FROM PRICE;

Price_Code	Price_Description	Price_RentFee	Price_DailyLateFee
1	Standard	2.50	1.00
2	New Release	4.00	3.00
3	Discount	2.00	1.00
4	Weekly Special	1.50	0.50

4 rows in set (0.001 sec)

Question 6

```
MariaDB [elitevideo]> SELECT Movie_Name, Movie_Year, Movie_Genre
FROM MOVIE;
```

Movie_Name	Movie_Year	Movie_Genre
The Cesar Family Christmas	2014	FAMILY
Smokey Mountain Wildlife	2011	ACTION
Richard Goodhope	2015	DRAMA
Beatnik Fever	2014	COMEDY
Constant Companion	2015	DRAMA
Where Hope Dies	2005	DRAMA
Time to Burn	2012	ACTION
What He Doesn't Know	2013	COMEDY

8 rows in set (0.000 sec)

Question 7

```
MariaDB [elitevideo]> SELECT Movie_Year, Movie_Name, Movie_Cost FROM
MOVIE ORDER BY Movie_Year DESC;
```

Movie_Year	Movie_Name	Movie_Cost
2015	Richard Goodhope	59.95
2015	Constant Companion	89.95
2014	The Cesar Family Christmas	39.95
2014	Beatnik Fever	29.95
2013	What He Doesn't Know	58.29
2012	Time to Burn	45.49
2011	Smokey Mountain Wildlife	59.95
2005	Where Hope Dies	25.49

8 rows in set (0.001 sec)

Question 8

```
MariaDB [elitevideo]> SELECT Movie_Name, Movie_Year, Movie_Genre
FROM MOVIE ORDER BY Movie_Genre ASC,Movie_Year DESC;
```

Movie_Name	Movie_Year	Movie_Genre
Time to Burn	2012	ACTION
Smokey Mountain Wildlife	2011	ACTION
Beatnik Fever	2014	COMEDY
What He Doesn't Know	2013	COMEDY
Richard Goodhope	2015	DRAMA
Constant Companion	2015	DRAMA
Where Hope Dies	2005	DRAMA
The Cesar Family Christmas	2014	FAMILY

8 rows in set (0.001 sec)

Question 9


```
MariaDB [elitevideo]> SELECT Movie_Num, Movie_Name, Price_Code FROM
MOVIE WHERE Movie_Name LIKE 'R%';
```

Movie_Num	Movie_Name	Price_Code
1236	Richard Goodhope	2

1 row in set (0.001 sec)

Question 10

```
MariaDB [elitevideo]> SELECT Movie_Name, Movie_Year, Movie_Cost FROM
MOVIE WHERE Movie_Name LIKE '%Hope%' ORDER BY Movie_Name;
```

Movie_Name	Movie_Year	Movie_Cost
Richard Goodhope	2015	59.95
Where Hope Dies	2005	25.49

2 rows in set (0.001 sec)

Question 11

```
MariaDB [elitevideo]> SELECT Movie_Name, Movie_Year, Movie_Genre
FROM MOVIE WHERE Movie_Genre = 'ACTION';
```

Movie_Name	Movie_Year	Movie_Genre
Smokey Mountain Wildlife	2011	ACTION
Time to Burn	2012	ACTION

2 rows in set (0.001 sec)

Question 12

```
MariaDB [elitevideo]> SELECT Movie_Num, Movie_Name, Movie_Cost FROM
MOVIE WHERE Movie_Cost > 40;
```

Movie_Num	Movie_Name	Movie_Cost
1235	Smokey Mountain Wildlife	59.95
1236	Richard Goodhope	59.95
1238	Constant Companion	89.95
1245	Time to Burn	45.49
1246	What He Doesn't Know	58.29

5 rows in set (0.001 sec)

Question 13

```
MariaDB [elitevideo]> SELECT Movie_Num,Movie_Name,Movie_Cost,
Movie_Genre FROM MOVIE WHERE Movie_Genre IN ('ACTION', 'COMEDY')AND
Movie_Cost < 50 ORDER BY Movie_Genre;
```

Movie_Num	Movie_Name	Movie_Cost	Movie_Genre
1245	Time to Burn	45.49	ACTION
1237	Beatnik Fever	29.95	COMEDY

2 rows in set (0.001 sec)

Question 14

```
MariaDB [elitevideo]> SELECT Mem_Num,Mem_FName,Mem_LName,
Mem_Street, Mem_State, Mem_Balance FROM MEMBERSHIP WHERE Mem_State =
'TN' AND Mem_Balance < 5 AND Mem_Street LIKE '%Avenue';
```

Mem_Num	Mem_FName	Mem_LName	Mem_Street	Mem_State	Mem_Balance
104	Jamal	Melendez	788 East 145th Avenue	TN	0.00
112	Luis	Trujillo	7267 Melvin Avenue	TN	3.00

2 rows in set (0.001 sec)

Question 15

```
MariaDB [elitevideo]> SELECT Movie_Genre,COUNT(*) AS Num_Movies FROM
MOVIE GROUP BY Movie_Genre;
```

Movie_Genre	Num_Movies
ACTION	2
COMEDY	2
DRAMA	3
FAMILY	1

4 rows in set (0.001 sec)

Question 16

```
MariaDB [elitevideo]> SELECT AVG(Movie_Cost) AS Average_Cost FROM
MOVIE;
```

Average_Cost
51.127500

1 row in set (0.002 sec)

Question 17

```
MariaDB [elitevideo]> SELECT Movie_Genre,AVG(Movie_Cost) AS  
Average_Cost FROM MOVIE GROUP BY Movie_Genre;
```

Movie_Genre	Average_Cost
ACTION	52.720000
COMEDY	44.120000
DRAMA	58.463333
FAMILY	39.950000

4 rows in set (0.002 sec)

Question 18

```
MariaDB [elitevideo]> SELECT M.Movie_Name,  
M.Movie_Genre,P.Price_Description,P.Price_RentFee FROM MOVIE M JOIN  
PRICE P ON M.Price_Code = P.Price_Code;
```

Movie_Name Price_RentFee	Movie_Genre	Price_Description
The Cesar Family Christmas 4.00	FAMILY	New Release
Smokey Mountain Wildlife 2.00	ACTION	Discount
Richard Goodhope 4.00	DRAMA	New Release
Beatnik Fever 4.00	COMEDY	New Release
Where Hope Dies 2.00	DRAMA	Discount
Time to Burn 2.00	ACTION	Discount
What He Doesn't Know 2.50	COMEDY	Standard

7 rows in set (0.001 sec)

Question 19

```
MariaDB [elitevideo]> SELECT M.Movie_Genre,AVG(P.Price_RentFee) AS  
Avg_Rental_Fee FROM MOVIE M JOIN PRICE P ON M.Price_Code =  
P.Price_Code GROUP BY M.Movie_Genre;
```

Movie_Genre	Avg_Rental_Fee
-------------	----------------

ACTION	2.000000
COMEDY	3.250000
DRAMA	3.000000
FAMILY	4.000000

4 rows in set (0.001 sec)

Question 20

```
MariaDB [elitevideo]> SELECT
M.Movie_Name,M.Movie_Cost,P.Price_RentFee,ROUND(M.Movie_Cost /
P.Price_RentFee,2) AS BreakEven_Rentals FROM MOVIE M JOIN PRICE P ON
M.Price_Code = P.Price_Code;
```

Movie_Name BreakEven_Rentals	Movie_Cost	Price_RentFee
The Cesar Family Christmas 9.99	39.95	4.00
Smokey Mountain Wildlife 29.98	59.95	2.00
Richard Goodhope 14.99	59.95	4.00
Beatnik Fever 7.49	29.95	4.00
Where Hope Dies 12.75	25.49	2.00
Time to Burn 22.75	45.49	2.00
What He Doesn't Know 23.32	58.29	2.50

7 rows in set (0.005 sec)

Question 21

```
MariaDB [elitevideo]> SELECT Movie_Name, Movie_Year FROM MOVIE WHERE
Price_Code IS NOT NULL;
```

Movie_Name	Movie_Year
What He Doesn't Know	2013
The Cesar Family Christmas	2014
Richard Goodhope	2015
Beatnik Fever	2014

Smokey Mountain Wildlife	2011	
Where Hope Dies	2005	
Time to Burn	2012	
+-----+		

7 rows in set (0.002 sec)

Question 22

MariaDB [elitevideo]> SELECT Movie_Name,Movie_Genre,Movie_Cost FROM MOVIE WHERE Movie_Cost BETWEEN 44.99 AND 49.99;

Movie_Name	Movie_Genre	Movie_Cost	
+-----+			
Time to Burn	ACTION	45.49	
+-----+			

1 row in set (0.001 sec)

Question 23

MariaDB [elitevideo]> SELECT M.Movie_Name, P.Price_Description, P.Price_RentFee FROM MOVIE M JOIN PRICE P ON M.Price_Code = P.Price_Code WHERE M.Movie_Genre IN ('FAMILY','COMEDY','DRAMA');

Movie_Name	Price_Description	Price_RentFee	
+-----+			
The Cesar Family Christmas	New Release	4.00	
Richard Goodhope	New Release	4.00	
Beatnik Fever	New Release	4.00	
Where Hope Dies	Discount	2.00	
What He Doesn't Know	Standard	2.50	
+-----+			

5 rows in set (0.001 sec)

Question 24

MariaDB [elitevideo]> SELECT DISTINCT M.Mem_Num, M.Mem_FName, M.Mem_LName, M.Mem_Balance FROM MEMBERSHIP M JOIN RENTAL R ON M.Mem_Num = R.Mem_Num;

Mem_Num	Mem_FName	Mem_LName	Mem_Balance	
+-----+				
102	Tami	Dawson	11.00	
103	Curt	Knight	6.00	
104	Jamal	Melendez	0.00	
105	Iva	McClain	15.00	
107	Rosario	Elliott	5.00	
110	Lewis	Rosales	0.00	
111	Stacy	Mann	8.00	
+-----+				

7 rows in set (0.001 sec)

Question 25

```
MariaDB [elitevideo]> SELECT MIN(M.Mem_Balance) AS  
Min_Balance,MAX(M.Mem_Balance) AS Max_Balance,AVG(M.Mem_Balance) AS  
Avg_Balance FROM MEMBERSHIP M JOIN RENTAL R ON M.Mem_Num =  
R.Mem_Num;
```

Min_Balance	Max_Balance	Avg_Balance
0.00	15.00	7.555556

1 row in set (0.000 sec)

Question 26

```
SELECT R.Rent_Num, R.Rent_Date,  
V.Vid_Num,Movie_Name,D.Detail_DueDate,D.Detail_ReturnDate FROM RENTAL R  
JOIN DETAILRENTAL D ON R.Rent_Num = D.Rent_Num JOIN VIDEO V ON D.Vid_Num  
= V.Vid_Num JOIN MOVIE M ON V.Movie_Num = M.Movie_Num WHERE  
D.Detail_ReturnDate > D.Detail_DueDate ORDER BY R.Rent_Num, M.Movie_Name;
```